

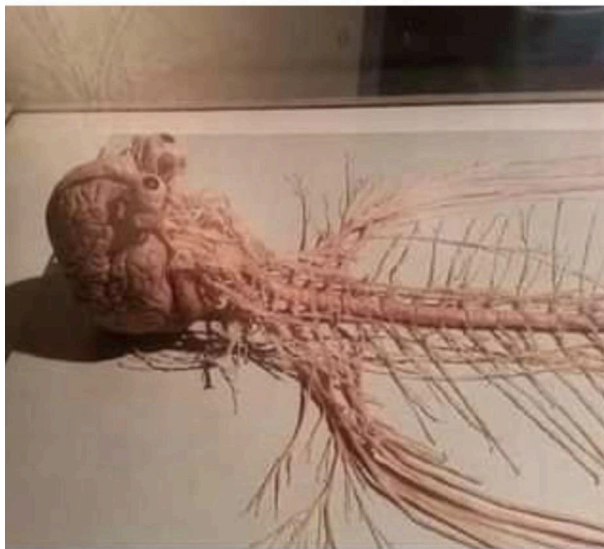
2025-2026 Hawk & Hornet Invitational

Anatomy & Physiology C

EXAM BOOKLET

**DO NOT WRITE
ON THESE PAGES!**

d-do u ever just



just get to thinkin'

Total Score: ____ / 138

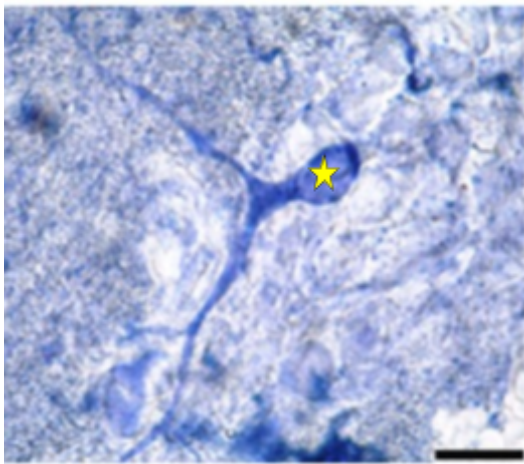
Instructions & Information

- 50 minutes will be given for the exam
- Each team may bring writing utensils, two Class II calculators, and one (1) 8.5" x 11" sheet of paper that may contain information on both sides in any form and from any source. The sheet of paper may be laminated or placed in a sheet protector to increase durability. Affixed labels, as well as multiple sheets of paper (whether in a single sheet protector or not) are prohibited.
- Tiebreakers will be denoted with **TB#** and will be in the following order: **3, 14, 21, 28, 60, 69, 85, 104**

Good luck!

I. Nervous System

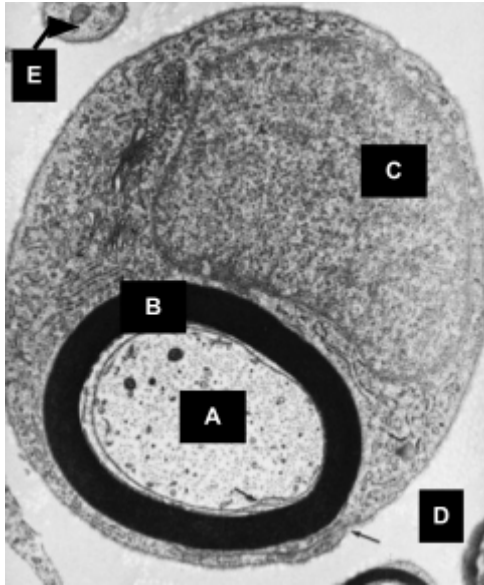
1. True (A) / False (B): All interneurons are found exclusively in the central nervous system.
2. True (A) / False (B): All Schwann cell-enveloped neurons are myelinated.
3. **[TB#1]** True (A) / False (B): Myelination increases the speed of signal conduction in neurons.
4. True (A) / False (B): Astrocytes are the most abundant glial cells in the CNS.
5. True (A) / False (B): White matter is named for the pearly appearance it has due to its high myelin content.



6. The starred area represents the:
 - a. Axon
 - b. Axon hillock
 - c. Dendrites
 - d. Soma
 - e. Varicosity
7. The histological section of the neuron referenced in the previous question depicts a(n):
 - a. Anaxonic neuron
 - b. Unipolar neuron
 - c. Bipolar neuron
 - d. Multipolar neuron
 - e. None of the above; it shows a junction between neurons
8. In a reflex arc, sensory neurons pass on information received most directly to _____.
 - a. Receptors
 - b. Afferent neurons
 - c. Efferent neurons
 - d. Interneurons
 - e. Effectors

9. What is the region located between the dura mater and arachnoid mater of the spinal cord?
 - a. Meningeal space
 - b. Epidural space
 - c. Subarachnoid space
 - d. Meninges
 - e. Denticulates
10. _____ are CNS glial cells responsible for producing and circulating cerebral spinal fluid.
 - a. Schwann cells
 - b. Satellite cells
 - c. Ependymal cells
 - d. Microglia
 - e. Astrocytes
11. With regards to myelin production, an analogy can be made that Schwann cells are to the PNS as _____ are to the CNS.
 - a. Oligodendrocytes
 - b. Microglia
 - c. Satellite cells
 - d. Ependymal cells
 - e. Astrocytes
12. This division of the nervous system is responsible for signalling to the skeletal muscles, resulting in voluntary and involuntary contractions and reflexes.
 - a. Central division
 - b. Visceral sensory division
 - c. Somatic sensory division
 - d. Visceral motor division
 - e. Somatic motor division
13. Local potentials, responsible for transmitting signals down an axon, differ from action potentials in all of the following ways except for:
 - a. Local potentials vary in magnitude along the strength of the stimulus
 - b. Local potentials become weaker as they travel from their origins
 - c. Local potentials are reversible
 - d. Local potentials can be both excitatory and inhibitory
 - e. All of the above statements are true
14. **[TB#2]** This molecule causes spastic paralysis, resulting in conditions such as lockjaw.
 - a. Curare
 - b. Botulinum toxin
 - c. Anatoxin-a
 - d. Tetanus toxin
 - e. Tetrodotoxin

15. _____ is the threshold potential of a neuron.
- 20 mV
 - 55 mV
 - 70 mV
 - 85 mV
 - 100 mV



The diagram above shows an electron micrograph of an axon (bisection). Which label above corresponds to:

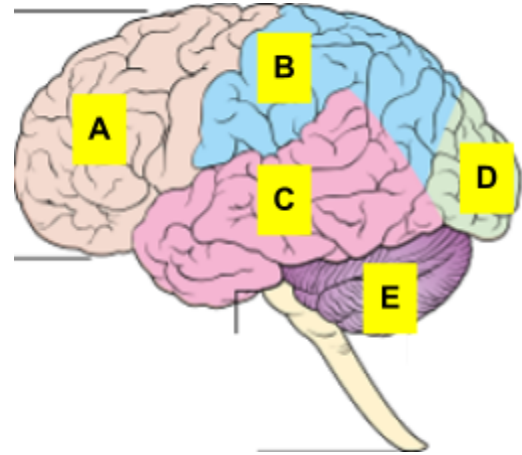
16. _____ The Schwann cell?
17. _____ The axon?
18. _____ The myelin sheath?

19. Some signals cross over from one side of the body to another; the optic chiasm is a well-known example of this. This phenomenon is known as:

- Involution
- Volution
- Regression
- Decussation
- Dissection

20. Which neurotransmitter is responsible for the “runner’s high” experienced by athletes?

- Epinephrine
- Norepinephrine
- Dopamine
- Enkephalins
- Beta-endorphin



21. **[TB#3]** Which of the regions shown is not considered a region within the forebrain?

- Region B
- Region D
- Region E
- B and C
- All of the above

22. Which region best corresponds to the parietal lobe?

- Region A
- Region B
- Region C
- Region D
- Region E

23. This region of the brain is heavily involved in processing signals from the eye and vision as a whole.

- Region A
- Region B
- Region C
- Region D
- All of the above

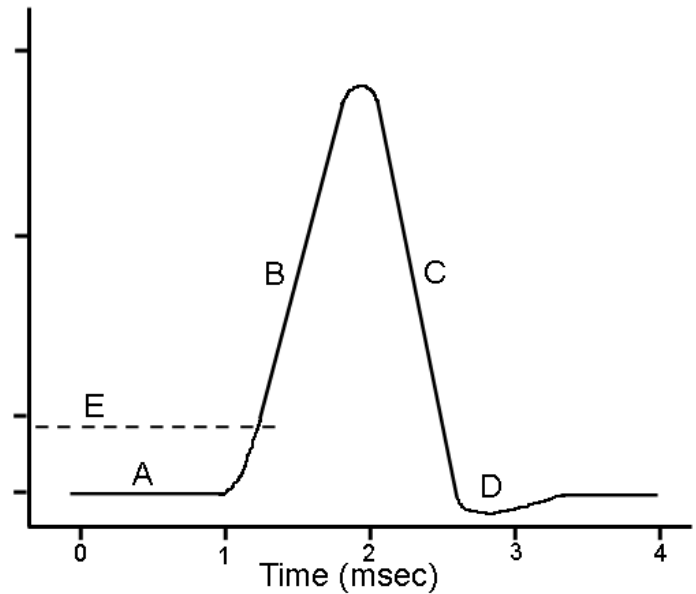
24. What are the primary function(s) of Region E?

- Short-term memory and smell
- Coordinating voluntary movements and posture
- Personality and decision making
- Speech and language comprehension
- Spatial processing

25. Sensory information is carried up the spinal cord towards the brain by:

- Ascending tracts
- Descending tracts
- Sympathetic tracts
- Parasympathetic tracts
- More than one of the above

26. This neurotoxin is derived from several fish as well as select bacteria. It blocks sodium channels and can often be lethal.
- Tubocurarine
 - Tetrodotoxin
 - Toxiferene
 - Anatoxin-a
 - Botulinum toxin
27. What reflex serves to protect oneself from loud sounds?
- Withdrawal reflex
 - Pupillary reflex
 - Tympanic reflex
 - Blink reflex
 - Knee-jerk reflex
28. **[TB#4]** Olfactory cells are best described as:
- Unipolar neurons
 - Bipolar neurons
 - Multipolar neurons
 - Anaxonic neurons
 - None of the above; olfactory cells are epithelial in nature
29. Which of the following component(s) of the eye are neural in nature?
- Conjunctiva
 - Uvea
 - Retina
 - Trochlea
 - More than one of the above
30. Spastic paralysis can be found in cases of poisoning with:
- Botulinum
 - Curare
 - Tetanus
 - A and C
 - All of the above
- For the following questions, select the label (A-E) above which BEST corresponds to the descriptions below. Some labels may be used more than once, while some may not be used at all.



- _____ Depolarization.
- _____ Hyperpolarization.
- _____ Is generally around -70 mV.
- _____ Differentiates an action potential from a graded-potential (all-or-none).
- _____ Rapid flow of sodium ions into a cell.
- _____ Inactivation of sodium channels, and opening of voltage-gated potassium channels.
- _____ Occurs because there is a slight delay in the closing of voltage-gated potassium channels.

Match the following statements with the neurotransmitter which is best described.

- | | |
|--|------------------|
| 38. _____ An inhibitory amino acid neurotransmitter common in the spine. | A. Acetylcholine |
| 39. _____ Is present at most autonomic nervous system synapses. Known for exciting skeletal muscle while inhibiting cardiac muscle. | B. Glutamate |
| 40. _____ The most common inhibitory neurotransmitter in the brain. Helps regulate stress. | C. Glycine |
| 41. _____ Present in the hypothalamus, cerebellum, retina, spinal cord. Regulates digestion and bowel function as well as temperature, alertness, mood, and sleepiness. | D. GABA |
| 42. _____ Is responsible for the majority of excitatory signals in the brain and the most common neurotransmitter in the brain. It is important for learning and memory. | E. Serotonin |

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Identify whether the following processes are associated with the sympathetic (A) or parasympathetic (B) nervous system.

43. _____ Epinephrine secretion.

45. _____ Constricting airways.

47. _____ Constricting pupils.

44. _____ Inhibition of intestine
activity.

46. _____ Increasing heart rate.

48. [2 pts] What does EPSP stand for in the context of the nervous system? What about IPSP?

49. What is a projection pathway?

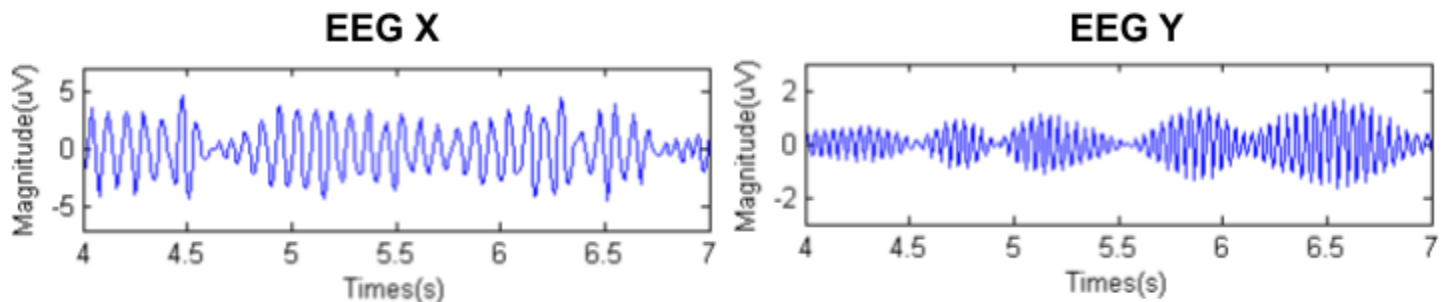
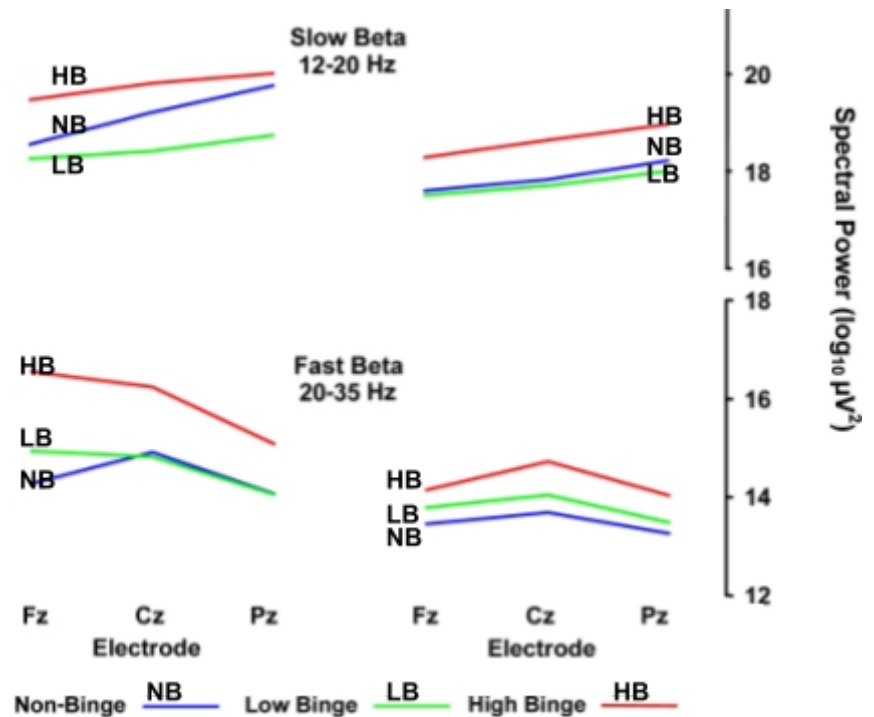
End of test.
Just kidding. There is more :)

Adapted from Courtney & Polich (2010). doi: 10.3390/ijerph7052325.

Binge drinking is a social phenomenon with a high prevalence among undergraduate students.

The major goal of the present study was to assess resting EEG in young adults who varied in their alcohol-drinking amount in the absence of a family history for alcoholism.

The fast-beta spectral power yielded a significant binge group main effect, and post-hoc assessment found that the high-binge group exhibited greater mean spectral power than the non-binge group (15.1 vs. 13.9 μV^2 , $p = 0.03$), with a marginal difference obtained between high- and low-binge drinkers (15.1 vs. 14.2 μV^2 , $p = 0.10$). These findings are consistent with those from previous studies reporting that alcoholics exhibit increased EEG spectral power in the beta band.



50. [4 pts] Of the waves shown in the figure, which one corresponds to an alpha wave? Which corresponds to a beta wave?
51. What physiological states are alpha waves associated with?
52. What physiological states are beta waves associated with?
53. Based on the results of the study, would you expect alcohol to have a greater stimulatory effect in high binge drinkers or non-binge drinkers?

The neurobiological mechanisms which mediate alcohol's stimulant effects are well understood. Researchers attribute these effects to activation of the brain's "reward circuitry" (Hendler et al., 2013). Alcohol intake is seen as a contributing, modifiable risk factor for strokes in young adults.

54. Identify the primary neurotransmitter associated with rewards.
55. [2 pts] Briefly, what is a stroke?
56. [2 pts] Identify one alcohol-related risk factor which contributes to stroke.

Special Senses

57. True (A) / False (B): All sensory signals are ultimately delivered to the CNS.
58. True (A) / False (B): If the skin is touched in two different locations within the same receptive field, a person will only sense one combined touch rather than two distinct touches.
59. True (A) / False (B): The bumps one can see and feel on their tongue are taste buds.
60. **[TB#5]** True (A) / False (B): Of all the tastes, the tongue is the most sensitive to bitter tastes.
61. True (A) / False (B): Olfactory neurons are the only neurons which remain exposed to the external environment.
62. Which of the following is not considered a special sense?
 - a. Vision
 - b. Hearing
 - c. Equilibrium
 - d. Smell
 - e. All of the above are special senses
63. What type of papillae is most abundant on the human tongue?
 - a. Filiform
 - b. Foliate
 - c. Fungiform
 - d. Vallate
 - e. All of the above are equally distributed
64. Molecules responsible for _____ flavor(s) rely on G-protein coupled receptors and second messenger systems for signal transduction.
 - a. Sweet
 - b. Salty
 - c. Bitter
 - d. A and B
 - e. A and C
65. Which of the following is not one of the primary tastes?
 - a. Salty
 - b. Sweet
 - c. Umami
 - d. Bitter
 - e. Spicy
66. Aside from the tongue, taste buds can be found in the:
 - a. Soft palate
 - b. Epiglottis
 - c. Inside the cheeks
 - d. A and C
 - e. All of the above
67. The taste of food is influenced by:
 - a. Texture
 - b. Temperature
 - c. Appearance
 - d. B and C
 - e. All of the above
68. Replacement of old olfactory cells occurs via division of:
 - a. Other olfactory cells
 - b. Basal cells
 - c. Granule cells
 - d. Mitral cells
 - e. Old olfactory cells cannot be replaced when irreparably damaged
69. **[TB#6]** From most external to internal, select the correct order of the following:
 - a. Incus → malleus → stapes
 - b. Incus → stapes → malleus
 - c. Malleus → incus → stapes
 - d. Stapes → malleus → incus
 - e. Stapes → incus → malleus
70. The incus, stapes, and malleus concentrate force from the _____ onto the _____.
 - a. Tympanic membrane; oval window
 - b. Tympanic membrane; round window
 - c. Oval window; round window
 - d. Oval window; tympanic membrane
 - e. Round window; oval window
71. What kind of ion channel is found at the tip of stereocilia?
 - a. Cl^-
 - b. Na^+
 - c. K^+
 - d. Ca^{2+}
 - e. Mg^{2+}

72. Which of the following statements regarding hair cells is false? Indicate whether the following structures are components of the outer (A), middle (B), or inner (C) ear.
- There are more than twice the number of outer hair cells as compared to inner hair cells
 - Outer hair cells have a greater number of stereocilia compared to inner hair cells
 - Inner hair cells are responsible for generating the signals which result in hearing
 - Outer hair cells are solely protective in function
 - None of the above; all these statements are true
73. _____ Auditory ossicles.
74. _____ Auricle.
75. _____ Vestibule.

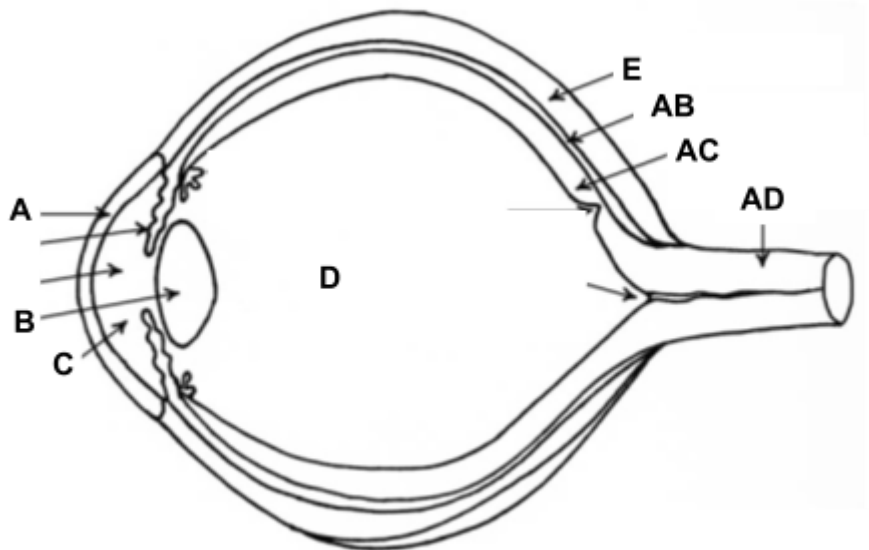
Match the following molecules to their corresponding tastes. Some tastes may be repeated, while some may not be used at all.

A. Salty B. Sour C. Bitter D. Sweet E. Umami

76. _____ Aspartic acid.
77. _____ Electrolytes such as potassium.
78. _____ Glutamate.
79. _____ H^+ ions.
80. _____ Caffeine.
81. _____ Solanine, theobromine, and other similar.

Match the following components with the eye shown.

82. _____ Aqueous humor.
83. _____ Vitreous humor.
84. _____ Sclera.
85. [TB#7] _____ Cornea.
86. _____ Optic nerve.

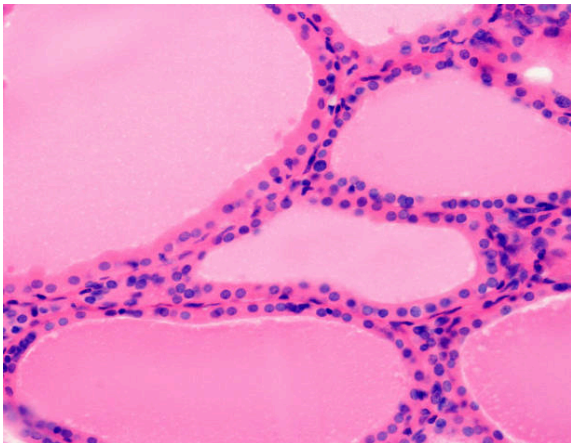


Fill in the following blanks with the appropriate terms.

87. [2 pts] The _____ is responsible for detecting linear acceleration on the horizontal plane, while the _____ is responsible for detecting linear acceleration on the vertical plane.
88. [2 pts] The structures from the previous question are housed in bulbs known as the _____ at the end of the _____. The latter structure is responsible for sensing rotational movements.
89. These calcium-carbonate and protein granules referred to as _____ are present to add weight and inertia to membranes responsible for sensing gravity and motion,
90. [3 pts] During _____ of the eye, the ciliary muscles _____, resulting in a(n) _____ of the curvature of the lenses of the eye and thus enhancing focus on nearby objects.
91. [3 pts] At the center of the retina is a small, oval area responsible for central vision known as the _____. Within this area is a pit known as the _____ with an exceptionally high density of _____ cells.
92. [2 pts] _____ occurs when the focal point is in front of the retina (near-sightedness, can occur when eyeball is too long). _____ describes the opposite (farsightedness, can occur when eyeball is too short).
93. [2 pts] _____ describes farsightedness which occurs with age. _____ is the medical term for having normal, 20/20 vision.

Endocrine System

94. True (A) / False (B): All neurotransmitters are hormones, but not all hormones are necessarily neurotransmitters.
95. True (A) / False (B): Goiter is associated with both Graves' disease and Hashimoto's disease.
96. True (A) / False (B): Endocrine glands typically release hormones directly into the bloodstream.
97. True (A) / False (B): The average adult typically has four parathyroid glands.
98. True (A) / False (B): All steroid hormones are derived from cholesterol.



99. Identify the organ displayed in the histological section above.

- Hypothalamus
 - Pituitary gland
 - Thyroid
 - Thymus
 - Adrenal gland
100. The primary, active hormone secreted by the hormone above is responsible for:
- Increasing heart rate
 - Increasing respiratory rate
 - Increasing heartbeat strength
 - A and B
 - All of the above
101. Glucagon is secreted from the _____.
- Hypothalamus
 - Thyroid
 - Pancreas
 - Adrenal glands
 - Thalamus

102. Which of the following hormone-organ/structure pairings is incorrect?
- PRL; mammary gland
 - TSH; thyroid
 - FSH; ovary
 - GH; liver
 - ACTH; kidneys
103. The adrenal cortex is known for producing _____ hormones.
- Amine
 - Peptide
 - Steroid
 - Nucleic acid
 - Glycoprotein
104. **[TB#8]** The pineal gland is most known for secreting:
- Oxytocin
 - Serotonin
 - Dopamine
 - Melatonin
 - Adrenaline
105. All of the following are examples of peptide hormones except for:
- Antidiuretic hormone
 - Growth hormone
 - Prolactin
 - Cortisol
 - Parathyroid hormone
106. Calcitonin is functionally an antagonist of which of the following hormones?
- PTH
 - TH
 - MSH
 - MT
 - INS
107. What type of cell is responsible for insulin secretion?
- Glomerulus cells
 - Spongiocytes
 - Clear cells
 - Beta cells
 - Parafollicular cells
108. During what time does growth hormone secretion generally peak?
- Midday
 - At night
 - During both mornings and evenings
 - During times of fasting
 - During times of stress

109. Identify the step in the cycle below which is incorrect.

0) TRH is released by the hypothalamus.

- 1) TRH stimulates TSH secretion.
- 2) TSH stimulates TH secretion.
- 3) TH increases metabolism of numerous cells in the body.
- 4) TH stimulates further TSH secretion.
- None of the steps above are incorrect.

110. The GH-IGF system is responsible for:

- Increasing protein synthesis within affected cells
- Increasing metabolism within affected cells
- Reducing glucose metabolism within affected cells
- A and B
- All of the above



111. **[TB#9]** Identify the term used to refer to the growth shown above.

- Gyrus
- Metastasis
- Ganglion
- Goiter
- Globus pallidus

112. Which of the following hormones exhibits positive feedback?

- Insulin
- Glucagon
- Parathyroid hormone
- Oxytocin
- A and B

113. ADH is also known as:

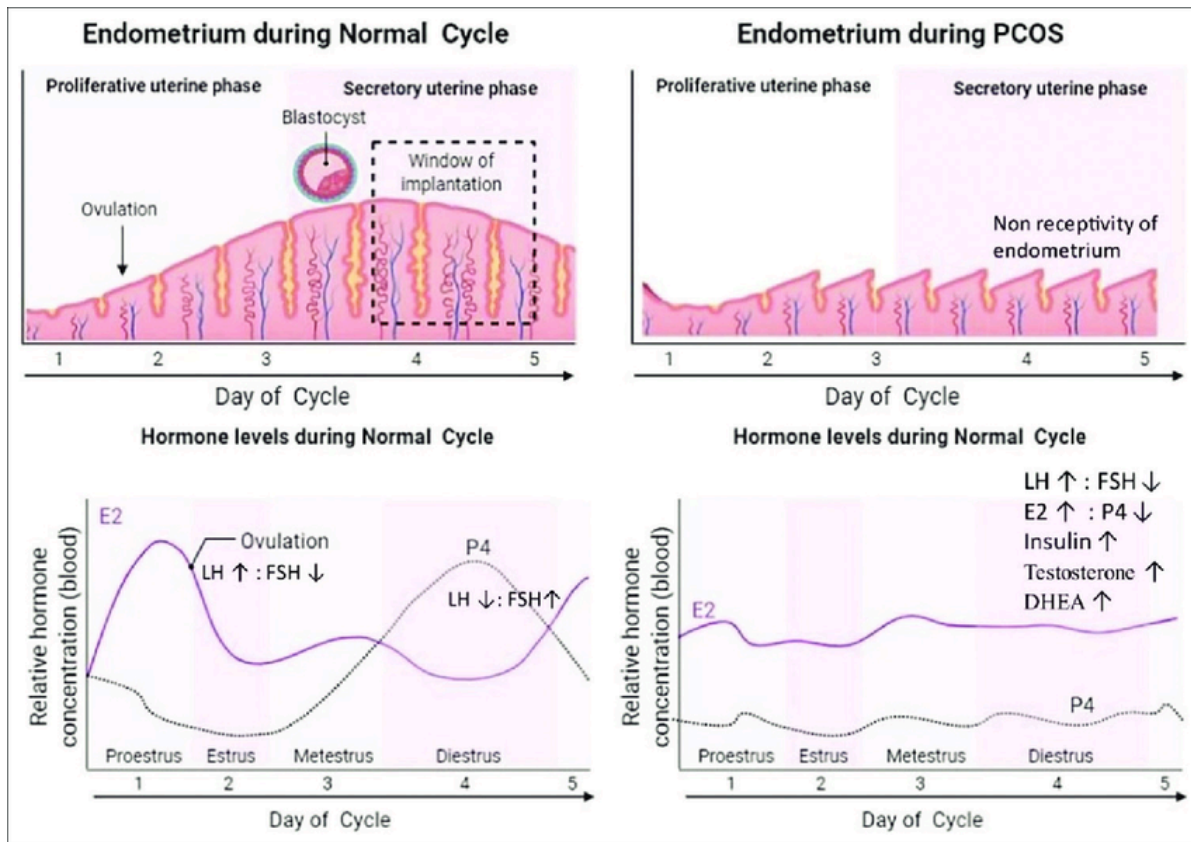
- Adrenocorticotrophic hormone
- Vasopressin
- Aldosterone
- Angiotensin
- Dehydroepiandrosterone

For the following hormones, identify whether they are secreted by the anterior pituitary (A), posterior pituitary (B), or hypothalamus.

114. _____ ADH.

115. _____ Follicle-stimulating hormone.

116. _____ Thyrotropin-releasing hormone.



The diagrams above indicate levels present for E2 (estradiol) and P4 (progesterone).

117. [2 pts] What is the primary difference between E2 and E3 estrogens?
118. Briefly describe the primary function of P4 in the menstrual cycle.
119. Identify the hormone released by the hypothalamus to increase LH and FSH secretions.
120. List the two primary phases of the menstrual cycle.
121. [2 pts] Although polycystic ovary syndrome affects both of the previous phases, issues generally arise from one phase in particular which affect the other. Identify which phase this is, and how PCOS begins to exert its effects.
122. Identify one diagnostic criteria of PCOS.